

Chethan Reddy

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AI Software Engineer (SDE-2) | AI Forward Deployed Engineer

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India

TECHNICAL EXPERIENCE

- **Staple AI** Singapore / Remote
AI Software Engineer (SDE-2); current remit: AI FDE May 2025 – Present
AI Software Engineer (SDE-1), SCANNING Team May 2024 – Jun 2025
 - Own end-to-end delivery for **20+ enterprise customers**: requirements, product engineering, backend integrations, QA automation, customer success and ongoing engineering support.
 - Develop scalable document AI for platform workflows processing **100k documents/month across 50+ enterprise customers**, including OCR and stamp, logo and signature detection.
 - Lead feature architecture, zero-shot LLMops, RAG analysis, debugging and optimization; review code, mentor engineers and integrate vision-language providers with consistent prompts, parsing and output normalization.
 - Implemented **OpenTelemetry** instrumentation for asynchronous workflows, recording execution duration, pages, retries, template matches and CPU/memory usage.
 - Built template-threshold APIs with validation and MongoDB persistence; improved table extraction, OCR-coordinate normalization, LLM-response guards and fallbacks; separated database read/write connections.
 - Work with Python, AWS, Kubernetes, Bastion hosts, OCR/LLM services and LangChain for prompt optimization, deployment and operational troubleshooting.
- **Dotnitrion Technologies / Alvarez & Marsal** Remote
AI Software Engineer, Dotnitrion Technologies Dec 2023 – May 2024
Alvarez & Marsal engagement, facilitated by Dotnitrion Dec 2023 – Jun 2024
 - Led Python backend development for **Diligence GPT** with Django, FastAPI and ArangoDB; designed RAG architecture with cross-functional teams and integrated LangChain/OpenAI into due-diligence workflows.
 - Led technical collaboration, reviewed code, debugged integrations and tested backend/AI features for software quality.
- **Bajaj Finserv Health** Pune / On-site
Data Science Engineer Intern, InSightRx Data Team May – Aug 2023
 - Led lab-test recommendation development, improving operational efficiency **30%**; applied LLMs, OCR, NER and YOLO to healthcare document/image analysis.
 - Optimized Elasticsearch/SQL retrieval with fuzzy matching and query refinement, improving retrieval speed **80%**.
- **TIFAC / Vellore Institute of Technology** Vellore / On-site
Project Intern, Backend Development and Data Dec 2022 – Jun 2023
 - Led **Physio Plus for MicroClinics** backend development with FastAPI, MongoDB and AWS distributed architecture, supporting patient-record analysis and **100+ daily appointments**; also worked on inventory-drone algorithms.
- **Smart Diet Planner** India
Machine Learning Researcher, Internship Aug 2022 – Apr 2023
 - Built search, automated PDF analysis, text/nutrition extraction and macro-consumption calculations for an AI diet application serving **850k+ users**; developed a customized LLM assistant for condition-specific diet recommendations.
- **Grroom** India
Machine Learning Intern Nov 2021 – Jan 2022
Team Lead Jan – Feb 2022
 - Trained **YOLO** models for fashion-image detection, tagging and annotation in Google Colab, completing **100k+ training iterations** and supporting backend integration; managed **10+ interns** building ML algorithms and Python scripts.

TECHNICAL SKILLS

Languages: Python, SQL, C++, Go, Java, JavaScript/TypeScript, HTML. **AI/data:** TensorFlow, Keras, NumPy, SciPy, pandas, Astropy, Plotly, LLMs, RAG, LangChain, NLP, OCR, NER, YOLO, computer vision, deep learning, regression, SVMs, fuzzy matching, geometric analysis. **Backend/web:** FastAPI, Django, Flask, Node/Express, Next.js, React, REST APIs. **Databases:** PostgreSQL, MySQL, SQLite, MongoDB, ArangoDB, Elasticsearch, NoSQL. **Infrastructure/quality:** AWS, Kubernetes, Google Colab, Linux, Git, OpenTelemetry, distributed architecture, code review, debugging, testing, QA automation, Postman, Render, VS Code. **Foundations:** data structures, algorithms, system design, operating systems, networking.

TEACHING, MENTORING AND SCIENTIFIC PROGRAMMING

- **RoboGems** Remote
Machine Learning Instructor, Part-time Aug 2022 – Jun 2023
 - Taught practical machine learning through personalized instruction; helped learners construct advanced deep-learning models and deploy completed models.
- **edvi** Delhi / Remote
Software Engineer Mentor, Freelance Oct 2022 – Mar 2023
 - Mentored learners in data science, ML, databases, system design, LLD/HLD, data structures, algorithms, computer networking and digital electronics, explaining implementation choices and solving technical problems.
- **Modo Edulabs** India
Machine Learning Content Associate, Part-time Aug 2022 – Mar 2023
 - Designed structured ML curricula with hands-on projects, revised multiple Python courses and trained learners from introductory through advanced machine learning.
- **Tekie** Bangalore
Python Programming Mentor, Part-time Jun 2021 – Oct 2022
 - Mentored **60+ students** through **500+ teaching hours** in Python/C++; debugged and evaluated assignments, problem-solving exercises and coding-test submissions.
 - Contributed to data-science/AI curriculum R&D; explained programming through visual demonstrations and structured lessons.
- **Camp K12** India
Coding Instructor, Python and AI, Part-time Jun – Aug 2022
 - Delivered **150+ hours** of Python, AI and ML instruction, with introductory blockchain content; personalized mentoring and designed projects around learners' interests.
- **Cybeorg Education Technology** Dubai / India
Coding Consultant Jun – Dec 2021
Coding Tutor Intern Mar – Jun 2021
 - Built advanced Python curricula covering programming mathematics, ML, data science, deep learning and astronomy; delivered **300+ hours** of instruction to school-age and adult learners.
 - Assisted a doctoral researcher with Python for astrophysics research using **NumPy, SciPy, pandas, TensorFlow, Astropy and Plotly**; developed practical projects and scientific-programming explanations.
- **CipherSchools** India
Teaching Assistant, Internship May – Jun 2021
 - Evaluated learners' Python scripts, projects and assignments; recommended improvements and maintained learner records using Excel and management systems.

TECHNICAL COMMUNITY AND RESEARCH TEAMS

- **VITrendz** Part-time
Machine Learning Team Apr 2022 – Jan 2023
Technical Team Member Aug 2021 – Apr 2022
 - Developed timetable algorithms and ML features for the student platform, including spam/inappropriate-comment filtering for faculty reviews; produced ML content, researched product improvements and mentored contributors.
- **Team AutoZ VIT** Part-time
Technical Team Member, Machine Learning Oct – Dec 2021
 - Researched autonomous-vehicle algorithms using YOLO object detection, Bayesian regression, neural-network regression and decision-forest regression.

EDUCATION

- **Vellore Institute of Technology** Vellore, India
Bachelor of Engineering, Electrical and Electronics Engineering
 - Participated in IEEE Computer Society, ABC group, Team AutoZ and TIFAC healthcare engineering.
- **Sree Vidyanikethan** — Higher Secondary Education, MPC; 93.4%.
- **Bharatiya Vidya Bhavan's** — High School; 90.1%.

RECENT PRODUCT ENGINEERING

- **Clinical Scribe / ScribeDesk:** Built a clinical scribe with **FastAPI, MongoDB, Next.js, TypeScript, Deepgram and OpenAI**: streaming transcription, editable notes, summaries, letters, encounter chat, PDF export and clinician sign-off. Improved transcript restoration, reconnects, session races and processor cleanup. In the subsequent collaborative **PostgreSQL/Redis** ScribeDesk implementation, contributed cookie fixes, startup migrations, worker configuration and monorepo deployment.
- **Salon Operations Platform:** Built booking, inventory, storefront, staff permissions, HR, payroll and finance with **Next.js, TypeScript, PostgreSQL and Prisma**, including commissions, loans, leave deductions, payslips and budget-versus-actual reports. Added an assistant over **17 allowlisted tables** with tokenized SQL validation, read-only transactions, statement limits and deterministic financial outputs. Cached SQL while refreshing query results; a local repeat-query check improved from **1,775 ms to 8 ms**. Implemented analytics, price propagation and consent-aware messaging infrastructure with signed webhooks, opt-outs and idempotent send claims.
- **AuditVault:** Built a maritime audit platform with **Go, Next.js, PostgreSQL and R2/S3**: eight-stage audit life-cycle, vessel-level access, mutation logs, resumable uploads and signed downloads; repaired cross-tenant authorization boundaries.
- **MarinePulse:** Built safety, maintenance, drill and engine-performance workflows; translated expert rules into explainable diagnostics. Implemented **121-item, 11-category** inspection scoring, AI-assistant tools and Word reports.
- **Sellos / RetailOS:** Developed retail POS with **Next.js, Go and PostgreSQL**: billing, inventory, expiry alerts, tax, reports, tenant schemas and row-locked stock updates. Contributed authentication, billing UX and deployment while integrating Subhanu's backend/admin work.
- **Applywise:** Built ATS discovery, explainable matching, profile ingestion, grounded résumé generation and Chrome form assistance with **React, Node/Express, SQLite, PostgreSQL/Neon, Chrome MV3 and LaTeX/Tectonic**. Implemented asynchronous drafting, independent review/correction, PDFs and resumable progress; bound delivery approval to exact job/answer/document versions with audited outcomes and duplicate prevention.
- **Websites and Product Exploration:** Built Brahmly's **15-page** educational website and Rogell Advisory's website with SEO, metadata, resources, contact flows and responsive layouts. Created CargoVision's accessible visual teaser and a Sustaino concept preview, plus social-publishing and administration scaffolds.

APPLIED AI AND SOFTWARE PROJECTS

- **eduAId — Instructor Analytics:** Built engagement/topic analytics with **over 90% accuracy**, multiple LangChain agents and handwritten-note digitization; implemented FastAPI/MongoDB services for teaching insights, summaries and learning support.
- **KaRmA — KRA-AI-Manager:** Built an employee-performance chatbot and report generator with **Mistral 7B, Llama, LangChain, SQL, custom agents/prompts and FastAPI**.
- **hAIr — AI in HR:** Implemented résumé parsing, keyword-based candidate ranking, customized interview questions and video-input evaluation within an automated hiring workflow.
- **Supplier Performance (Aug 2022):** Ranked suppliers across **7m+ records** using rating, cost, resources and delivery time; used MongoDB, polynomial regression and Plotly for **five-year** forecasts. [Code](#).
- **Keywords Recommendation Engine (Jun 2022):** Built insurance-search autocomplete, autocorrect and recommendations with **linear SVM, Levenshtein, Fast Autocomplete and DAGs**, delivering searches in **under 800 ms**. [Code](#).
- **WeDio (Apr 2022):** Applied geometric analysis and Deep Vision to **five HD-video disruption classes**: lag, blur, quality defects, black screens and freezing; generated Excel/JSON timestamps, types and scores. [Code](#).
- **FFCS Planner Auto Slot Picker (Nov 2021):** Optimized timetables across **2,000+ courses** using course, faculty and time preferences and **3D graph traversal**, supporting **10,000+ VIT students**. [Code](#).
- **PhotoBook:** Built a Python/FastAPI API with **Pillow and FPDF** to turn photographs and page dimensions into downloadable PDF layouts. [Code](#).

HACKATHON ACHIEVEMENTS

- **Honeywell / IEEE WIE VIT WeHack 3.0 — 1st Place, Apr 2022:** CCTV video analytics.
- **Bajaj Finserv HackRx 3.0 — 1st Runner-up, Jun 2022:** Search and recommendation systems.
- **Axis Bankathon 2023 — 1st Runner-up:** LLM-based employee KRA assistant.
- **Bitget U-30 2023 — Most Viable Project award.**
- **Canara Bank Dacoethon 2024 — Top 5 of 3,000+ teams.**
- **GLC-HP Hackathon 2024 — Grand Finalist / Top 4 of 3,000+ teams.**
- **Sirion Labs HackFest 1.0 — Top 7 of 2,000+ teams, Aug 2022:** Grand finalist; supplier optimization.
- **Samsung Solve for Tomorrow 2022 — Top 50 of 18,000+ teams:** Phase 2; AmbuFast ambulance concept.
- **Siemens Healthineers Shift — Finalist, Nov 2022:** X-ray computer vision.
- **Amadeus H4S — Top 5:** Travel advisor using 3D graphs.
- **Make-a-thon — Top 10:** AI Sight smartphone eye-disease analysis.
- **SmartIdeathon 2023 — Top 100:** Multimodal transportation.
- **AI Arena 1.0 — Top 10:** Prompt-driven data analytics.
- **Walmart Sparkathon — Top 7:** Supplier analytics with LLMs.
- **Honeywell Hackwell 4.0 — Top 10:** Neural-network approaches to airport taxiing.

CERTIFICATIONS AND PROGRAM CREDENTIALS

- [Cricket Code Champions Hack](#) — HackerEarth, Dec 2023.
- [Hack The Mountains 3.0](#) — Hack The Mountains, Aug 2022.
- [Goldman Sachs Engineering Virtual Program](#) — Goldman Sachs, May 2022.
- [Transfer Learning for NLP with TensorFlow Hub](#) — Coursera, Jan 2021.
- [Programming for Everybody: Getting Started with Python](#) — Coursera, Oct 2020.
- [Python Data Structures](#) — Coursera, Oct 2020.
- [Data Science Math Skills](#) — Coursera, Oct 2020.
- [Introduction to Solar Cells](#) — Coursera, Oct 2020.
- [Introduction to Calculus](#) — Coursera, Oct 2020.
- [The Science of Success: What Researchers Know That You Should Know](#) — Coursera, Nov 2020.
- [Crash Course on Python](#) — Coursera, Nov 2020.
- [Electric Power Systems](#) — Coursera, Nov 2020.
- [Introduction to Sustainability](#) — Coursera, Nov 2020.
- [Using Python to Access Web Data](#) — Coursera, Nov 2020.
- [Vector Calculus for Engineers](#) — Coursera, Nov 2020.
- [Computer Vision Basics](#) — Coursera, Nov 2020.
- [Introduction to Electronics](#) — Georgia Tech via Coursera, Nov 2020.
- [Getting Started with AWS Machine Learning](#) — Coursera, Nov 2020.
- [Code Yourself! An Introduction to Programming](#) — Coursera, Nov 2020.